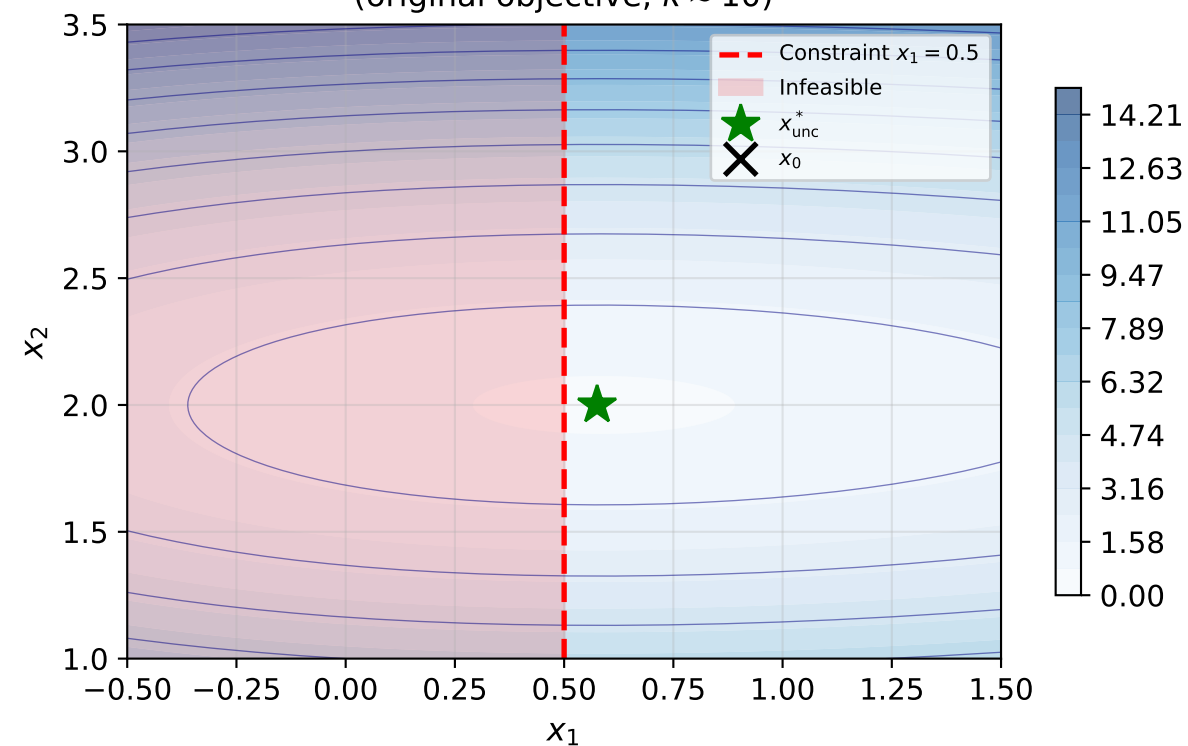


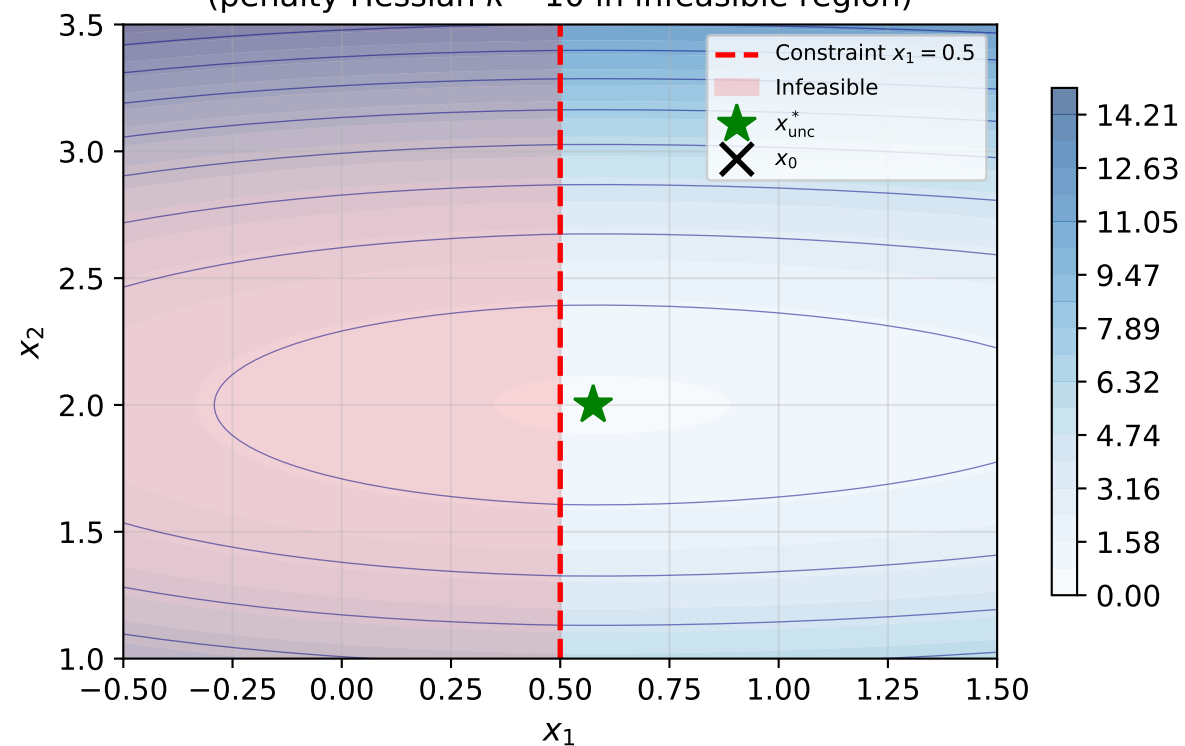
Q5 Penalty Landscape for Different λ Values

Large λ creates steep walls (ill-conditioning), requiring smaller step sizes

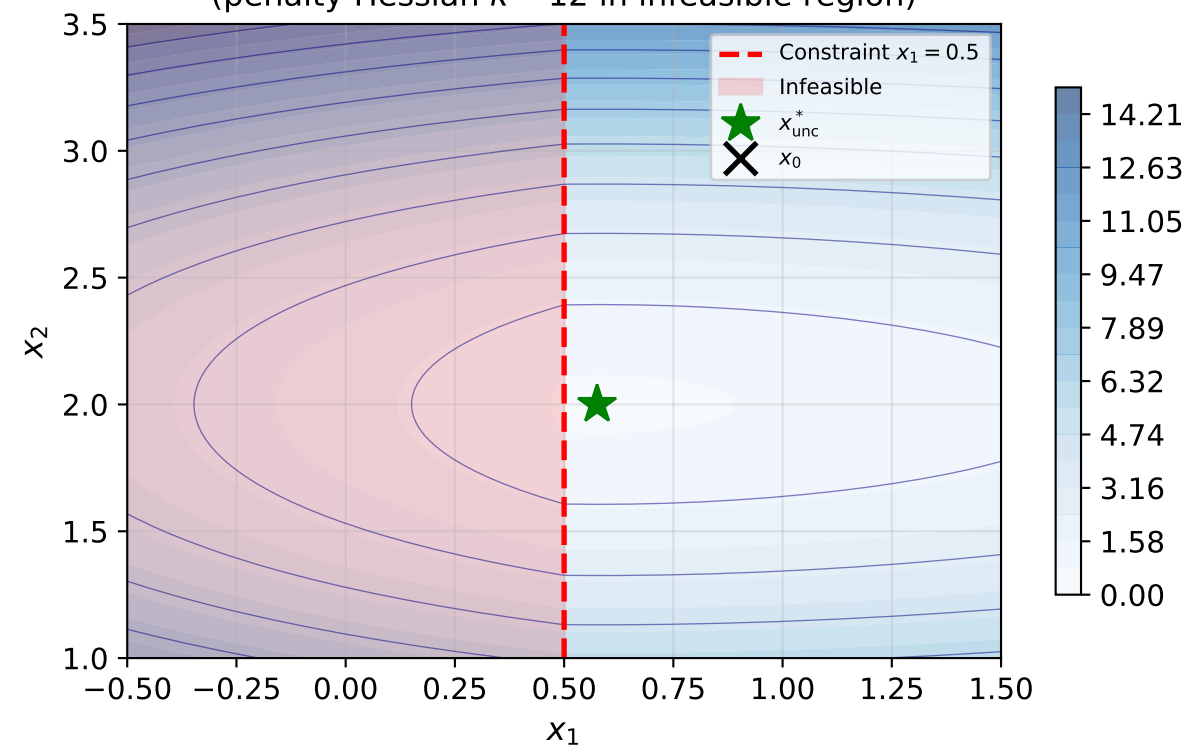
Unconstrained
(original objective, $\kappa \approx 10$)



$\lambda = 0.15$
(penalty Hessian $\kappa \sim 10$ in infeasible region)



$\lambda = 1.8$
(penalty Hessian $\kappa \sim 12$ in infeasible region)



$\lambda = 4.5$
(penalty Hessian $\kappa \sim 14$ in infeasible region)

